IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-023

1. Topic (12 words max)

ROAD INSPECTOR – Road Quality Inspection And Repair Estimation System

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

The quality of road infrastructure is essential for public safety, economic activity, and efficient transportation. In Sri Lanka and many developing regions, road inspections are mainly conducted manually by field officers, making the process slow, inconsistent, and highly dependent on human judgment. As a result, early-stage defects such as cracks, potholes, surface erosion, and poor tar application are often missed. The groups most affected by this problem include road authorities, municipal councils, construction companies, and daily road users such as drivers, pedestrians, and public transport operators.

Poor road condition assessment has serious social, economic, and technical impacts. Socially, damaged roads increase accident risks and reduce mobility. Economically, late detection leads to higher repair costs, vehicle damage, and disruptions to transport and trade. Technically, the lack of automated and data-driven inspection systems prevents accurate monitoring and predictive maintenance. Environmentally, inefficient repair planning causes material wastage and unnecessary carbon emissions, while ethically and legally, unsafe roads can result in loss of life and reduced public trust in authorities.

This research belongs to the **Technology Integration and Management (TIM)** cluster, as it integrates image processing, sensor data, and environmental analysis to support intelligent infrastructure management. The related industry sectors include transportation, civil engineering, smart infrastructure, and public services.

The proposed **ROAD INSPECTOR** system directly contributes to the **United Nations Sustainable Development Goals (SDGs)**, particularly **SDG 9 (Industry, Innovation, and Infrastructure)** by promoting resilient and sustainable infrastructure, **SDG 11 (Sustainable Cities and Communities)** through safer and more reliable road networks, and **SDG 13 (Climate Action)** by reducing unnecessary material usage and emissions through accurate repair estimation.

References *:*

1. I. H. Abbas & M. Q. Ismael, *Automated Pavement Distress Detection Using Image Processing Techniques*, Engineering, Technology & Applied Science Research, Oct 2021.
Automated Pavement Distress Detection

2. S. Sivanraj, D.N.L.S. Uduwage, R.K. Shiwakoti & M. Thripathi, *YOLOv11-s-Based Pothole Detection Model with Integrated Traffic Data for Maintenance Decision Support*, Journal of Advanced College of Engineering and Management, 2025.
https://www.nepjol.info/index.php/JACEM/article/view/84529

3. *Automated Road Defect and Anomaly Detection for Traffic Safety: A Systematic Review*, MDPI Sensors, 2023.
https://www.mdpi.com/1424-8220/23/12/5656

6. Brief description of Existing Research and Systems (300 words max)

Existing road inspection systems mainly rely on manual field surveys and visual assessments conducted by engineers and road officers. While some authorities use digital tools such as GPS mapping, cameras, and pavement management systems, defect identification and severity assessment are still largely dependent on human judgment. Advanced commercial road survey systems exist, but they require specialized vehicles and high investment, making them less practical for developing countries such as Sri Lanka.

Recent research has increasingly applied computer vision and deep learning techniques for road defect detection. Models such as Convolutional Neural Networks (CNNs), YOLO, and U-Net have been widely used to detect potholes, cracks, and surface irregularities from road images captured using smartphones, drones, or vehicle-mounted cameras. Some studies also estimate defect size and surface damage using image segmentation and depth analysis.

However, most existing systems focus only on detecting or classifying defects and do not provide complete maintenance decision support. They rarely integrate repair area calculation, material estimation, and repair time prediction into a single platform. In addition, many models are trained on datasets from developed countries, making them less accurate for Sri Lankan road conditions, which vary in surface material, lighting, and damage patterns. Environmental and traffic conditions are also often ignored.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution, **ROAD INSPECTOR**, is an automated road quality inspection and repair estimation system designed for road networks in Sri Lanka and similar developing regions. The system uses image processing, sensor data, and environmental analysis to detect road defects, measure surface damage, estimate repair area, calculate material requirements, and predict repair time. It addresses the limitations of existing manual and semi-automated road inspection methods, which are slow, inconsistent, and highly dependent on human judgment. The architecture is modular and integrated, allowing different components to operate independently while contributing to a unified infrastructure management platform.

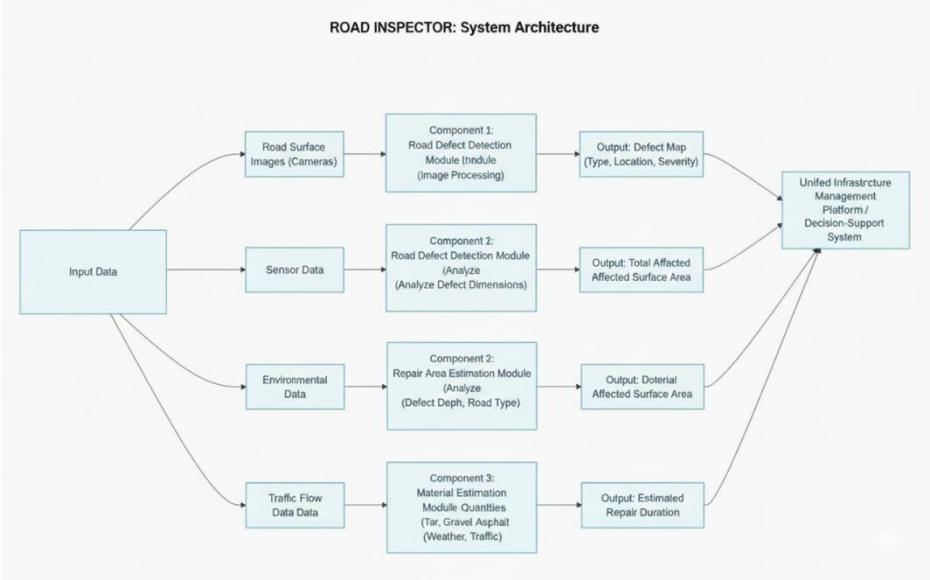
The system consists of four main functional components, each focusing on a key aspect of road maintenance while working together as a complete decision-support system.

Component 1 – Road Defect Detection Module
This component captures road surface images using mounted or handheld cameras and applies image processing techniques to identify visible defects such as potholes, cracks, surface erosion, and tar quality issues. The output is a defect map that highlights the type, location, and severity of each detected problem. This module replaces subjective manual inspection with a consistent and objective digital assessment.

Component 2 – Repair Area Estimation Module
This component analyzes the detected defects to calculate the total affected surface area. By measuring the dimensions and spread of each defect, the system determines how much of the road requires repair. This information is critical for planning maintenance work and avoiding under- or over-estimation of repair needs.

Component 3 – Material Estimation Module
Based on the estimated repair area, defect depth, and road type, this module calculates the quantity of materials required, such as tar, gravel, or asphalt. This allows maintenance teams to prepare accurate budgets and reduce material wastage, which is common in traditional repair planning.

Component 4 – Environmental and Repair Time Analysis Module
This component integrates weather conditions, traffic flow, and surface characteristics to estimate how long repairs will take. By considering these external factors, the system supports better scheduling and minimizes disruption to road users.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

To ensure the successful development of the ROAD INSPECTOR system, the project team will adopt a multi-disciplinary approach to bridge the gaps between civil engineering, computer vision, and data management. Specialized domain expertise in road construction and maintenance will be acquired through direct consultation with industry professionals and road authorities to ensure that defect classifications—such as differentiating between fatigue cracks and structural erosion—align with official engineering standards. This is supplemented by a rigorous study of material density charts and pavement management manuals to provide the necessary groundwork for the Material Estimation Module. On the technical front, the team will develop the required skills in image processing and sensor integration through structured training in Python-based libraries like OpenCV and TensorFlow. This will be paired with iterative hardware prototyping to master the calibration of level sensors and microcontrollers for real-time data processing.

The data required for the project will be gathered through a hybrid collection strategy. Primary data will be captured using the system's proprietary camera modules on local road segments to ensure the AI models are trained on regional surface types and environmental conditions. To accelerate the learning process, the team will also utilize secondary data from open-source global road damage datasets for initial model pre-training. Finally, this project does require ethical clearance, primarily due to data privacy concerns. Since the system captures high-resolution imagery in public spaces, automated anonymization protocols will be implemented to blur human faces and vehicle license plates. Furthermore, formal permissions will be obtained from the relevant municipal or transport authorities before conducting field tests to ensure that the data collection process complies with local safety and privacy regulations.

9. Objectives and Novelty

Main Objective To develop an intelligent image-processing-based inspection system capable of detecting road defects and providing accurate repair estimation.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22082756	Identify road defects	*Capture road surface images using camera modules*	• *Preprocess images (noise removal, enhancement)* • *Automated multi-defect detection using image processing instead of manual inspection* • *Apply image processing techniques to detect potholes, cracks, uneven surfaces, and tar defects*
IT22207968	Analyze environmental and traffic conditions	Collect environmental data such as weather conditions	• Analyze traffic density and movement patterns • Study the impact of environmental and traffic

			conditions on repair duration
IT22252340	Estimate repair area	Measure defect dimensions from processed images	• Calculate affected surface area using geometric and pixel-based methods • Estimate repair zones accurately for different defect types • Validate repair area estimation with sample data
IT22113122	Calculate material requirements	Classify road type (asphalt, concrete, tar surface) to select appropriate repair materials.	• Automated material quantity estimation directly linked to image-processed defect data. • Eliminates manual calculations and reduces material wastage. • Provides accurate, data-driven material estimation based on defect dimensions and road type. • Supports cost-effective and optimized resource allocation for road maintenance.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Mr	Buddika Harshanath		11/01/2026
Co-Supervisor	Prof	Samantha Rajapaksha		
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				